

Engineering Electromagnetics II

Homework #3 Solutions

78 points total

ECE 332 – Fall 2025

Due Tuesday, November 25, 2025

1. A 200 MHz left-hand circularly polarized plane wave with an electric field modulus of 5 V/m is normally incident in air on a dielectric medium with $\epsilon_r = 4$ and occupies the region defined by $z \geq 0$. (20 points total)
 - (a) Write an expression for the electric field phasor of the incident wave given that the field is a positive maximum at $z = 0$ and $t = 0$. (5 points)

The angular frequency is $\omega = 2\pi(2 \times 10^8) \text{ rad/s} = 4\pi \times 10^8 \text{ rad/s}$ and the wavenumber in Medium 1 (air) is

$$k_1 = \frac{\omega}{c} = \frac{4\pi \times 10^8 \text{ rad/s}}{3 \times 10^8 \text{ m/s}} = \frac{4\pi}{3} \text{ rad/m}.$$

The incident wave is LHC polarized, which means the phasor of the electric field has the form

$$\tilde{\mathbf{E}}^i = a_0(\hat{\mathbf{x}} + e^{j\pi/2}\hat{\mathbf{y}})e^{-jk_1z} = a_0(\hat{\mathbf{x}} + j\hat{\mathbf{y}})e^{-jk_1z}$$

and

$$\mathbf{E}^i(z, t) = a_0 \cos(\omega t - kz)\hat{\mathbf{x}} - a_0 \sin(\omega t - kz)\hat{\mathbf{y}},$$

where

$$|\mathbf{E}^i| = \sqrt{a_0^2 \cos^2(\omega t - kz) + a_0^2 \sin^2(\omega t - kz)} = a_0 = 5 \text{ V/m}.$$

So, the electric field phasor is

$$\tilde{\mathbf{E}}^i = 5(\hat{\mathbf{x}} + j\hat{\mathbf{y}})e^{-j4\pi z/3} \text{ V/m.}$$

- (b) Calculate the reflection and transmission coefficients. (5 points)

The intrinsic impedance of Medium 1 (air) is $\eta_1 = \eta_0 = 120\pi \Omega$ and the intrinsic impedance of Medium 2 is

$$\eta_2 = \frac{\eta_0}{\sqrt{\epsilon_r}} = \frac{1}{2}\eta_0 = 60\pi \Omega.$$

So, the reflection coefficient is

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = \frac{-60\pi}{180\pi} = -\frac{1}{3}$$

and the transmission coefficient is

$$\tau = 1 + \Gamma = \frac{2}{3}.$$

- (c) Write expressions for the electric field phasors of the reflected wave, the transmitted waves, and the total field in the region $z \leq 0$. (5 points)

The wavenumber in Medium 2 is

$$k_2 = \frac{\omega}{u_{p2}} = \frac{\omega\sqrt{\epsilon_{r2}}}{c} = 2k_1 = \frac{8\pi}{3} \text{ rad/m.}$$

Therefore,

$$\tilde{\mathbf{E}}^r = \Gamma a_0(\hat{\mathbf{x}} + j\hat{\mathbf{y}})e^{jk_1 z} = -\frac{5}{3}(\hat{\mathbf{x}} + j\hat{\mathbf{y}})e^{j4\pi z/3} \text{ V/m,}$$

$$\tilde{\mathbf{E}}^t = \tau a_0(\hat{\mathbf{x}} + j\hat{\mathbf{y}})e^{-jk_2 z} = \frac{10}{3}(\hat{\mathbf{x}} + j\hat{\mathbf{y}})e^{-j8\pi z/3} \text{ V/m,}$$

$$\tilde{\mathbf{E}}_1 = \tilde{\mathbf{E}}^i + \tilde{\mathbf{E}}^r = 5(\hat{\mathbf{x}} + j\hat{\mathbf{y}}) \left(e^{-j4\pi z/3} - \frac{1}{3}e^{j4\pi z/3} \right).$$

- (d) Determine the percentages of the incident average power reflected by the boundary and transmitted into the second medium. (5 points)

The percentage of reflected average power is

$$100|\Gamma|^2 = 100/9 = 11.1\%$$

and the percentage of transmitted average power is

$$100|\tau|^2 \frac{\eta_1}{\eta_2} = 100 \left(\frac{4}{9} \right) \left(\frac{120\pi}{60\pi} \right) = 88.9\%.$$

2. Repeat the above problem but replace the dielectric medium with a poor conductor characterized by $\epsilon_r = 2.25$, $\mu_r = 1$, and $\sigma = 10^{-4}$ S/m (20 points total)

- (a) As in Problem 1, for Medium 1,

$$\eta_1 = \eta_0 = 120\pi \Omega, \quad k_1 = \frac{\omega}{c} = \frac{4\pi}{3}$$

and the electric field phasor for the incident wave is

$$\tilde{\mathbf{E}}^i = 5(\hat{\mathbf{x}} + j\hat{\mathbf{y}})e^{-j4\pi z/3} \text{ V/m}.$$

- (b) For Medium 2,

$$\frac{\sigma_2}{\omega\epsilon_2} = \frac{10^{-4}(36\pi)}{(4\pi \times 10^8)(2.25 \times 10^{-9})} = 4 \times 10^{-3}.$$

So, Medium 2 is a low-loss dielectric, which means

$$\alpha_2 \approx \frac{\sigma_2}{2} \sqrt{\frac{\mu_2}{\epsilon_2}} = \frac{\sigma_2}{2} \frac{\eta_0}{\sqrt{\epsilon_{r2}}} = \frac{10^{-4}120\pi}{2\sqrt{2.25}} = 1.26 \times 10^{-2} \text{ Np/m},$$

$$\beta_2 \approx \omega \sqrt{\mu_2 \epsilon_2} = \frac{\omega \sqrt{\epsilon_{r2}}}{c} = 2\pi \text{ rad/m},$$

$$\eta_2 \approx \sqrt{\frac{\mu_2}{\epsilon_2}} \left(1 + j \frac{\sigma_2}{2\omega\epsilon_2} \right) = \frac{120\pi}{\sqrt{2.25}} \left(1 + j(2 \times 10^{-3}) \right) \approx 80\pi \Omega.$$

The reflection coefficient is

$$\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = \frac{-40\pi}{200\pi} = -\frac{1}{5}$$

and the reflection coefficient is

$$\tau = 1 + \Gamma = \frac{4}{5}.$$

(c) In this case,

$$\tilde{\mathbf{E}}^r = \Gamma a_0(\hat{\mathbf{x}} + j\hat{\mathbf{y}})e^{jk_1 z} = -(\hat{\mathbf{x}} + j\hat{\mathbf{y}})e^{j4\pi z/3} \text{ V/m},$$

$$\tilde{\mathbf{E}}^r = \tau a_0(\hat{\mathbf{x}} + j\hat{\mathbf{y}})e^{-\alpha_2 z}e^{-j\beta_2 z} = 4(\hat{\mathbf{x}} + j\hat{\mathbf{y}})e^{-1.26 \times 10^{-2} z}e^{-j2\pi z} \text{ V/m},$$

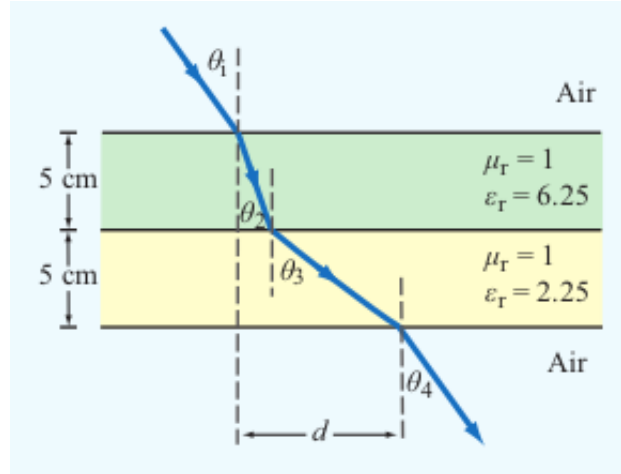
$$\tilde{\mathbf{E}}_1 = \tilde{\mathbf{E}}^i + \tilde{\mathbf{E}}^r = 5(\hat{\mathbf{x}} + j\hat{\mathbf{y}}) \left(e^{-j4\pi z/3} - \frac{1}{5}e^{j4\pi z/3} \right).$$

(d) The percentage of reflected average power is

$$100|\Gamma|^2 = 100/25 = 4\%$$

and the percentage of transmitted average power is

$$100|\tau|^2 \frac{\eta_1}{\eta_2} = 100 \left(\frac{16}{25} \right) \left(\frac{120\pi}{80\pi} \right) = 96\%.$$



3. A parallel-polarized plane wave is incident from air at an angle $\theta_i = 30^\circ$ onto a pair of dielectric layers, as shown above. (10 points total)

(a) Determine the angles of transmission θ_2 , θ_3 , and θ_4 . (5 points)

Snell's law of refraction implies

$$\sin \theta_2 = \sin \theta_i \sqrt{\frac{\epsilon_{r1}}{\epsilon_{r2}}} = \sin 30^\circ \sqrt{\frac{1}{6.25}} = 0.2.$$

So,

$$\theta_2 = \sin^{-1} 0.2 = 0.2014 \text{ rad} = 11.54^\circ.$$

Similarly,

$$\sin \theta_3 = \sin \theta_2 \sqrt{\frac{\epsilon_{r2}}{\epsilon_{r3}}} = 0.2 \sqrt{\frac{6.25}{2.25}} = \frac{1}{5} \frac{5}{3} = \frac{1}{3}.$$

Hence,

$$\theta_3 = \sin^{-1} \frac{1}{3} = 0.3398 \text{ rad} = 19.47^\circ.$$

Finally,

$$\sin \theta_4 = \sin \theta_3 \sqrt{\frac{\epsilon_{r3}}{\epsilon_{r4}}} = \frac{1}{3} \sqrt{\frac{2.25}{1}} = \frac{1}{3} \left(\frac{3}{2} \right) = \frac{1}{2}.$$

Therefore,

$$\theta_4 = \sin^{-1} \frac{1}{2} = \frac{\pi}{6} \text{ rad} = 30^\circ = \theta_i.$$

- (b) Determine the lateral distance d . (5 points)

The lateral distance that the wave travels is

$$d = (5 \text{ cm}) \tan \theta_2 + (5 \text{ cm}) \tan \theta_3 = 5(0.5577) \text{ cm} = 2.788 \text{ cm}.$$

4. A TE wave propagating in a dielectric-filled waveguide of unknown permittivity has dimensions $a = 5 \text{ cm}$ and $b = 3 \text{ cm}$. If the x component of its electric field is given by

$$E_x = -36 \cos(40\pi x) \sin(100\pi y) \sin\left((2.4\pi \times 10^{10})t - 52.9\pi z\right) \text{ (V/m)},$$

determine: (20 points total)

- (a) the mode number, (5 points) From the argument of the cosine factor,

$$\frac{m\pi}{a} = 40\pi \implies m = 40a = 40(0.05) = 2$$

and, from the argument of the spatial sine factor

$$\frac{n\pi}{b} = 100\pi \implies n = 100b = 100(0.03) = 3.$$

So, the mode is TE_{23} .

(b) ϵ_r of the material in the guide, (5 points)

From the sinusoidal time-dependent term,

$$\omega = 2.4\pi \times 10^{10} \text{ rad/s} \text{ and } \beta = 52.9\pi \text{ rad/m.}$$

So, using a formula from Example 8-8,

$$\begin{aligned} \epsilon_r &= \frac{c^2}{\omega^2} \left(\beta^2 + \left(\frac{m\pi}{a} \right)^2 + \left(\frac{n\pi}{b} \right)^2 \right) \\ &= \left(\frac{3 \times 10^8}{2.4\pi \times 10^{10}} \right)^2 \left((52.9\pi)^2 + (40\pi)^2 + (100\pi)^2 \right) \\ &= 2.25. \end{aligned}$$

(c) the cutoff frequency, (5 points)

For this medium,

$$u_{p0} = \frac{c}{\sqrt{\epsilon_r}} = \frac{3 \times 10^8 \text{ m/s}}{\sqrt{2.25}} = 2 \times 10^8 \text{ m/s.}$$

So, the cutoff frequency for this mode is

$$f_{23} = \frac{u_{p0}}{2} \sqrt{\left(\frac{2}{a} \right)^2 + \left(\frac{3}{b} \right)^2} = 10.77 \text{ GHz.}$$

(d) and the expression for H_y . (5 points)

Here,

$$Z_{\text{TE}} = \frac{E_x}{H_y} = \frac{\eta}{\sqrt{1 - (f_{23}/f)^2}} = \frac{377/\sqrt{\epsilon_r}}{\sqrt{1 - (10.77/12)^2}} = 569.9 \text{ } \Omega.$$

This means

$$H_y = \frac{E_x}{Z_{\text{TE}}} = -0.063 \cos(40\pi x) \sin(100\pi y) \sin((2.4\pi \times 10^{10})t - 52.9\pi z) \text{ A/m.}$$

5. A narrow rectangular pulse superimposed on a carrier with a frequency of 9.5 GHz was used to excite all possible modes in a hollow guide with $a = 3 \text{ cm}$ and $b = 2 \text{ cm}$. If the guide is 100 m in length, how long

will it take each of the excited modes to arrive at the receiving end?
(8 points)

The carrier can only excite a mode if its frequency is higher than the cutoff frequency for the mode. The modes with the lowest cutoff frequencies are the modes with lower values of m and n and a mode with higher values for m and n has a higher cutoff frequency.

Since the guide is “hollow”, the interior medium has the characteristics of free space, which means $u_{p0} = c$ and the cutoff frequencies for the lowest modes are

$$\begin{aligned} f_{10} &= \frac{c}{2a} = \frac{3 \times 10^8 \text{ m/s}}{2(3 \times 10^{-2} \text{ m})} = 5 \times 10^9 \text{ Hz} = 5 \text{ GHz} \\ f_{01} &= \frac{c}{2b} = \frac{3 \times 10^8 \text{ m/s}}{2(2 \times 10^{-2} \text{ m})} = 7.5 \times 10^9 \text{ Hz} = 7.5 \text{ GHz} \\ f_{11} &= \frac{c}{2} \sqrt{\frac{1}{a^2} + \frac{1}{b^2}} = (1.5 \times 10^8 \text{ m}) \sqrt{\frac{(4 + 9) \times 10^{-4}}{36 \times 10^{-8}}} \\ &= 1.5 \sqrt{\frac{13}{36}} \times 10^{10} = 9.01 \text{ GHz} \\ f_{20} &= \frac{c}{a} = 2f_{10} = 10 \text{ GHz}. \end{aligned}$$

All other modes have higher cutoff frequencies. So, the carrier excites TE_{10} , TE_{01} , TE_{11} , and TM_{11} . The group velocity of a mode is

$$u_g = c \sqrt{1 - (f_{mn}/f)^2}.$$

Using the cutoff frequencies calculated above for the excited modes,

$$u_g = \begin{cases} 0.85c = 2.55 \times 10^8 \text{ m/s}, & \text{TE}_{10} \\ 0.61c = 1.84 \times 10^8 \text{ m/s}, & \text{TE}_{01} \\ 0.32c = 0.95 \times 10^8 \text{ m/s}, & \text{TE}_{11} \text{ and } \text{TM}_{11}. \end{cases}$$

Therefore, the travel times for these modes are

$$u_g = \frac{d}{u_g} = \frac{100 \text{ m}}{u_g} = \begin{cases} 0.39 \text{ } \mu\text{s}, & \text{TE}_{10} \\ 0.54 \text{ } \mu\text{s}, & \text{TE}_{01} \\ 1.05 \text{ } \mu\text{s}, & \text{TE}_{11} \text{ and } \text{TM}_{11}. \end{cases}$$